

CYBERSECURITY INCIDENT RESPONSE REPORT

NIST SP 800-61 Lifecycle Structure

Attribute	Details
Incident ID	INCIDENT12345
Prepared By	Prakash Karki
Role	Cybersecurity Analyst
Program	Cyber Security Job-Ready Program
Target System	DESKTOP-2N2ATE
Data Source	Windows Security Event Logs (windows_logs.csv) analyzed in Splunk on Kali Linux
Report Date	September 2026

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1. Executive Summary

Analysis of the provided Windows Security Event Log identified a concentrated series of failed authentication attempts against host DESKTOP-2N2ATE. The evidence shows 62 Event ID 4625 failures within approximately 15 minutes on 17 June 2026, targeting 24 unique usernames. The local ADMINISTRATOR account was targeted 40 times (64.5% of the recorded failures). Three prominent external source IPs were identified from India, Latvia, and Brazil.

The observed activity is consistent with automated password guessing / password-spraying behavior. The supplied analysis did not identify an Event ID 4624 successful logon from the identified attacking IP addresses during the analyzed period. The source notes therefore support evidence of attempted unauthorized access, but do not establish a successful compromise.

Business impact:

- Potential exposure of a Windows host and privileged/local accounts to automated authentication attacks.
- Operational risk from repeated authentication failures, including possible account lockouts or service authentication disruption.
- No confirmed successful authentication or data compromise is documented in the supplied evidence.

Key takeaways:

- The attack concentrated heavily on the local ADMINISTRATOR account.
- The host appears to have been reachable through network-based authentication services; the original notes associate the activity with exposed administrative interfaces.
- Defensive priorities are to remove unnecessary public exposure, strengthen identity controls, improve rate-based detection, and validate monitoring coverage.

2. Incident Overview & Metadata

Attribute	Details
Incident ID	INCIDENT12345
Incident Type	Password Spraying / Brute Force Attack
Target Host	DESKTOP-2N2ATE
Affected Systems	DESKTOP-2N2ATE
Activity Start	2026-06-17 22:38:22 UTC
Activity End	2026-06-17 22:53:49 UTC
Log Volume	62 failed logon events
Unique Usernames	24
Unique Source IPs	10
Primary Target	ADMINISTRATOR — 40 of 62 attempts
Logon Type	Type 3 — Network
Primary Evidence	Windows Event ID 4625 records analyzed through Splunk

2.1 Scope and Evidence Limitations

- The report is based on the supplied CSV-derived Windows log analysis and the findings documented in the source report.
- The source notes state that no Event ID 4624 successful logon was observed from the identified malicious IPs during the analyzed period.
- A complete enterprise-wide compromise determination requires additional evidence such as endpoint telemetry, firewall/network logs, authentication logs, and EDR data.

3. Detection & Analysis Phase

3.1 Initial Detection / Detection Mechanism

The investigation identified an unexpected concentration of Windows Event ID 4625 (failed logon) events. The logs were imported into Splunk and aggregated by event ID, machine, username, source IP, and time. This enabled identification of repeated authentication attempts from multiple external IPs.

3.2 Initial Attack Vector

The activity used network-based authentication attempts (Logon Type 3) against the target Windows host. The supplied report associates this exposure with remote administrative interfaces and recommends removing direct public exposure of RDP/SMB.

3.3 Scope of Observed Activity

Attribute	Details
Total failed logons	62
Time period	2026-06-17 22:38:22–22:53:49 UTC
Unique usernames targeted	24
Unique source IPs	10
ADMINISTRATOR attempts	40 / 62 (64.5%)
103.165.78.91	27 attempts; geolocated in source notes to India; described as suspicious but clean in the cited VirusTotal check
185.118.79.30	19 attempts; geolocated in source notes to Latvia; two malicious reports noted
45.238.132.70	8 attempts; geolocated in source notes to Brazil; one malicious report noted

3.4 Analysis Timeline

Attribute	Details
22:38:22	Start of observed failed-logon activity.
22:38:22–22:53:49	62 Event ID 4625 failures recorded across 24 usernames and 10 source IPs.
During window	ADMINISTRATOR was targeted in 40 attempts.
22:53:49	Final recorded failure involved the Cerberus account; source notes interpret this as an internal service authentication failure occurring after the external activity.
Post-analysis	Splunk review documented no Event ID 4624 successful logon from the identified attacking IPs during the analyzed period.

3.5 Technical Indicators of Compromise

Attribute	Details
IP Address	103.165.78.91
IP Address	185.118.79.30
IP Address	45.238.132.70
Event ID	4625 — failed logon
Logon Type	3 — network logon
Target Account	ADMINISTRATOR
Other Targeted Accounts	ACCOUNTS, BACKUP, ROOT, ALEX, STUDENT, CERBERUS, and other usernames in the 24-account set

The Screenshots below showing the spunk search result of windows log.csv file for visualization.

- A. Checking if there is event id 4625 (**failed logons**) exist, and it noted that there are 62 logon attempts failed.

The screenshot shows the Splunk Enterprise interface with a search query: `index=windows | stats count AS Attempts by Username | sort -Attempts | head 5`. The search results show 62 events. The statistics table displays the following data:

Username	Attempts
ADMINISTRATOR	34
Administrator	6
8	1
????????	1
ACCOUNTS	1

A tooltip message states: "Enable event sampling to run the search and return a random set of events."

B. Checking how many unique IP were tried to log on into the system

The screenshot shows the Splunk Enterprise interface with a search query: `index= windows Id=4625 Source_IP!="-" | stats dc(Source_IP) AS Unique_IP`. The search results show 62 events. The statistics table displays the following data:

Unique_IP
11

C. Checking how many different user name tried by hacker

The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `index=windows | stats count AS Attempts by Username | sort -Attempts | head 5`. The search results show 62 events. The table displays the top 5 usernames and their attempt counts.

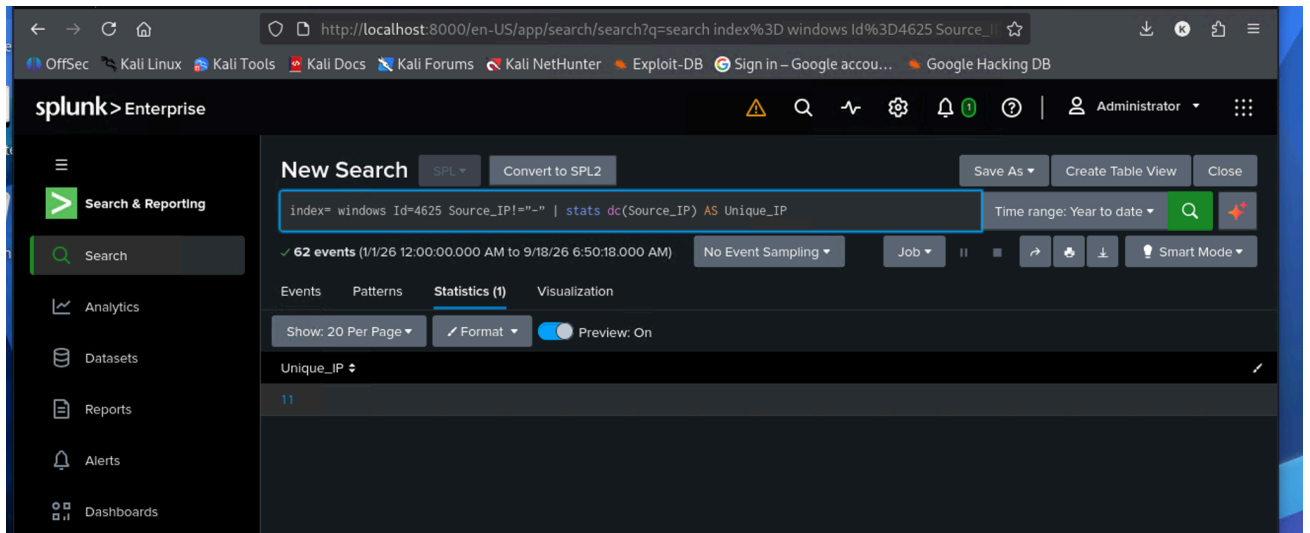
Username	Attempts
ADMINISTRATOR	34
Administrator	6
8	1
????????	1
ACCOUNTS	1

D. Checking how top 5 source IP repeated to attempt into the system

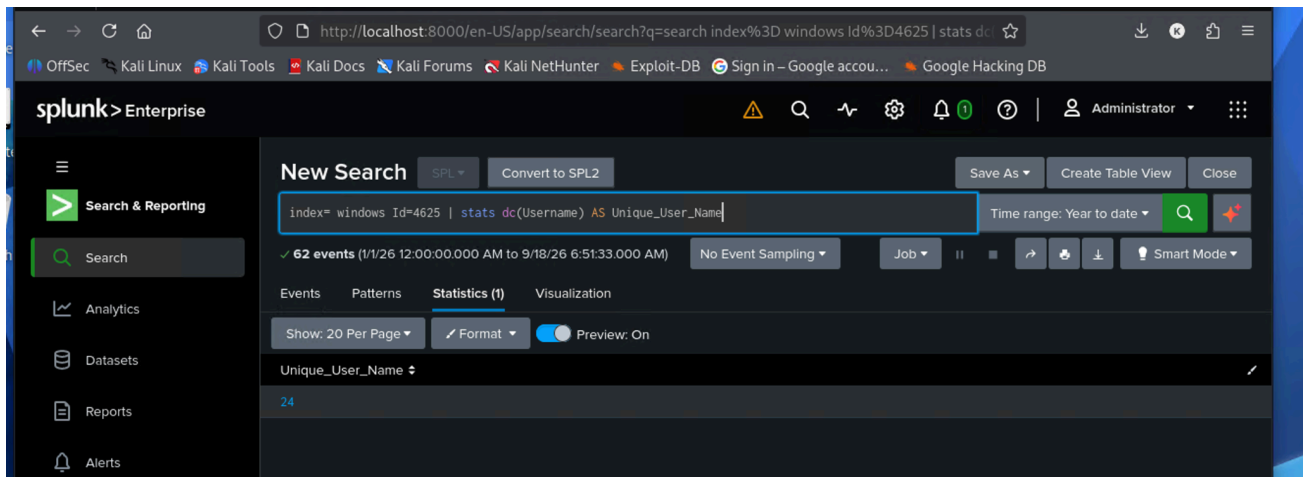
The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `index= windows Id=4625 Source_IP!="" | stats count AS Attempts by Source_IP | sort -Attempts | head 5`. The search results show 62 events. The table displays the top 5 source IP addresses and their attempt counts.

Source_IP	Attempts
103.165.78.91	27
185.118.79.30	19
45.238.132.70	8
-	1
118.26.111.63	1

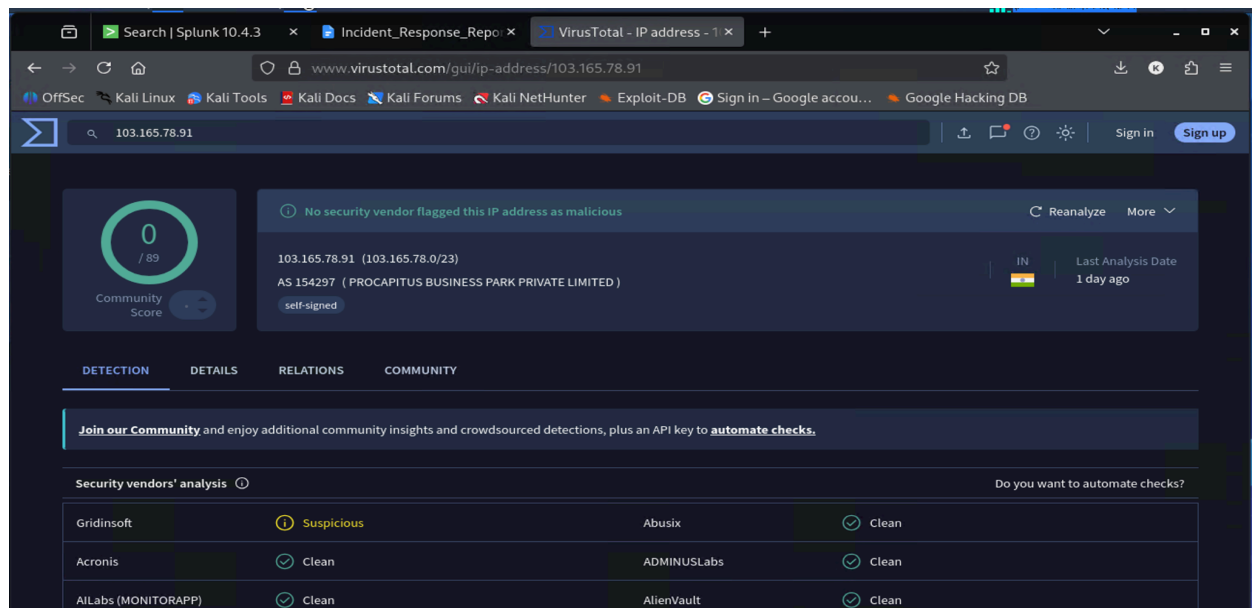
E. Checking how many distinct source IPs are used to log into the system.



- F. Checking how many unique user names attempt to login into the system.



- G. Checking top3 repeatedly tried IP to login into the system and its geolocation. Also checking if the IP address is reported as malicious by using www.virustotal.com
- It is found that IP address 103.165.78.91 tried to login from India
 - It is found to be clean but it is still suspicious.



- It is found that IP address 45.238.132.70 tried to login from Brazil
- There was one report that it was malicious.

4. Containment, Eradication, and Recovery

4.1 Short-Term Containment

- Block identified offending IP addresses at the appropriate perimeter firewall / network security control, subject to operational validation.
- Temporarily lock or protect high-risk accounts under active attack and initiate credential review.
- Monitor for continued Event ID 4625 activity from other source IPs to detect attacker rotation or distributed attempts.

4.2 Eradication / Permanent Remediation

- Remove direct public internet exposure of administrative interfaces such as RDP and SMB where not required.
- Require remote administrative access through an authenticated VPN or Zero Trust Network Access gateway with MFA.
- Audit local and domain accounts for weak, shared, default, or predictable credentials.

- Disable or rename built-in administrative accounts where operationally appropriate and deploy LAPS for managed local administrator credentials.
- Enforce strong password and account-lockout controls through centrally managed policy.
- Improve endpoint telemetry by deploying Sysmon and forwarding relevant process/network events to Splunk.

4.3 Recovery and Validation

- Review authentication logs to confirm whether any successful Event ID 4624 logon occurred from identified or related source IPs.
- Confirm that account-lockout and authentication protections are active and functioning as designed.
- Validate that administrative ports are no longer directly exposed to the public internet.
- Confirm normal operation of affected services, including any service accounts that experienced authentication failures.
- Maintain heightened monitoring after remediation to detect recurrence.

5. Post-Incident Activity / Lessons Learned

5.1 Root Cause Analysis

The source report identifies direct public internet exposure of DESKTOP-2N2ATE without sufficient edge firewall filtering or Network Level Authentication as the principal root cause. Based on the supplied evidence, this should be treated as the documented root-cause hypothesis until infrastructure and configuration records independently confirm the exposure path.

5.2 Contributing Factors

- Use of a predictable/high-value local Administrator account name.
- Insufficient restriction of network-based authentication exposure.
- Need for stronger rate-based alerting and automated SOC escalation.
- Limited telemetry in the supplied dataset for determining process-level activity or broader endpoint compromise.

5.3 Preventative Recommendations

Attribute	Details
Immediate	Enforce account lockout controls and maintain validated firewall blocking of identified malicious sources.
Medium-term	Restrict administrative interfaces; enforce NLA; require MFA-protected VPN for remote administration; onboard Sysmon telemetry into Splunk.
Detection	Create a real-time Splunk alert for more than 10 failed logons from a single IP within 60 seconds, with SOC escalation.
Identity	Disable/rename built-in local Administrator accounts where appropriate; deploy LAPS; enforce strong credential policy.
Long-term	Adopt Zero Trust Network Access / Zero Trust principles and phishing-resistant MFA such as FIDO2/WebAuthn for administrative access.

5.4 Playbook and Process Updates

1. Update the brute-force/password-spraying playbook to include Event ID 4625 triage by source IP, username, time window, and logon type.
2. Add a decision point for checking Event ID 4624 and other authentication success events before declaring the incident contained.
3. Add steps for validating public exposure of RDP/SMB and other authentication interfaces.
4. Define evidence-preservation requirements and minimum fields for incident timelines.
5. Document containment approval, change control, and rollback procedures for firewall and account actions.
6. Establish a post-containment monitoring period and explicit closure criteria.

5.5 Lessons Learned

- High-volume failed authentication activity can be identified efficiently by aggregating Event ID 4625 events by source IP and username.
- Privileged/default account names are attractive targets for automated credential attacks and should receive additional protection.
- Preventive architecture controls are more durable than relying only on IP blocklists, because attackers can rotate source infrastructure.
- Detection quality depends on correlated endpoint, authentication, network, and SIEM telemetry.

References

7. National Institute of Standards and Technology (NIST), SP 800-61 Rev. 2, Computer Security Incident Handling Guide.
8. MITRE ATT&CK, T1110 — Brute Force, including password guessing and password spraying techniques.
9. Microsoft Learn, Windows Security Auditing — Event ID 4625: An account failed to log on.
10. VirusTotal, threat intelligence and IP reputation service (as cited in the source report).
11. Splunk Documentation, Search Processing Language (SPL) and Search Reference.